

Types of Databases

Database Type	Description	Examples
Relational Database (RDBMS)	Stores data in tables with rows and columns. Uses SQL.	MySQL, PostgreSQL, Oracle
NoSQL Database	Non-relational. Flexible, scalable, schema-less.	MongoDB, Cassandra, CouchDB
Key-Value Store	Stores data as key-value pairs. Fast and scalable.	Redis, DynamoDB, Riak
Document Database	Stores data as documents (JSON/BSON).	MongoDB, Couchbase
Column-Oriented Database	Stores data in columns. Fast for analytics.	Apache Cassandra, HBase

Types of Databases (cont.)

Database Type	Description	Examples
Graph Database	Represents data as nodes and edges.	Neo4j, ArangoDB
Time-Series Database	Optimized for time-stamped data.	InfluxDB, TimescaleDB
Object-Oriented Database	Stores data as objects (OOP style).	db4o, ObjectDB
Hierarchical Database	Tree-like data structure.	IBM IMS, Windows Registry
Network Database	Allows many-to-many relationships.	IDS, TurboIMAGE

Types of Databases (cont.)

Database Type	Description	Examples
Multimodel Database	Supports multiple data models.	ArangoDB, OrientDB
NewSQL Database	Scalable SQL-based systems.	Google Spanner, CockroachDB
Cloud Database	Hosted in the cloud.	Amazon RDS, Firebase
Distributed Database	Data across multiple locations.	Cassandra, Couchbase
In-Memory Database	Stores data in RAM.	Redis, Memcached